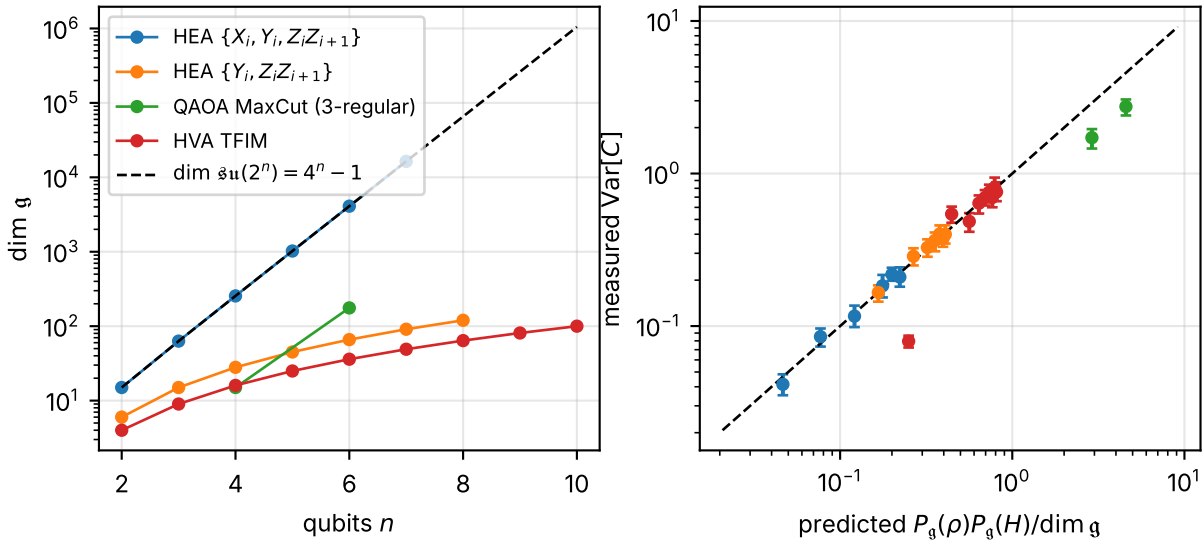
(a) dynamical Lie algebra dimension
HEA $\{X_i, Y_i, Z_iZ_{i+1}\}$
HEA $\{Y_i, Z_iZ_{i+1}\}$
QAOA MaxCut (3-regular)
HVA TFIM
dim $\mathfrak{su}(2^n) = 4^n - 1$
dim $\mathfrak{g}$
qubits n
(b) Lie-algebraic prediction vs. simulation
measured Var[C]
predicted $P_{\mathfrak{g}}(\rho)P_{\mathfrak{g}}(H)/\dim \mathfrak{g}$